

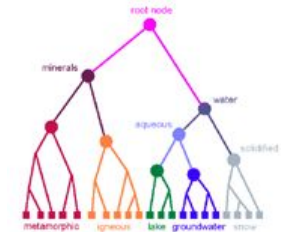
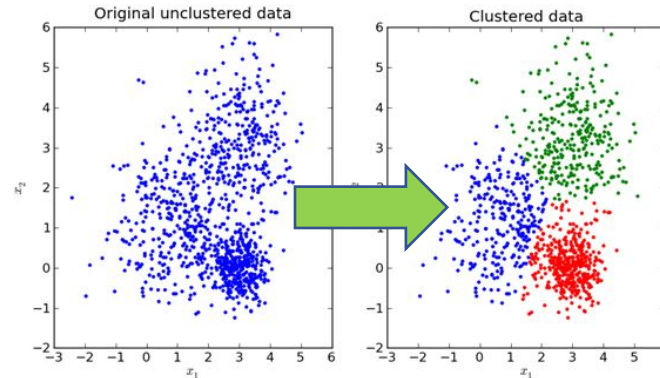
Problem Solving using Pattern Recognition

Day 4: Cluster Analysis

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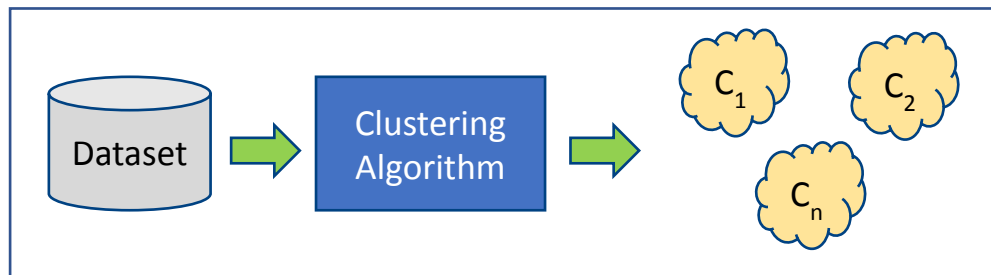


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1. Introduction to Cluster Analysis
2. Application of Cluster Analysis
3. Types of Clustering Methods
4. How to Validate the Created Clusters
5. How to Profile Clusters
6. Limitations of Cluster Analysis
7. Demonstration of clustering using software tools (e.g. Python)

1. Introduction to Clustering

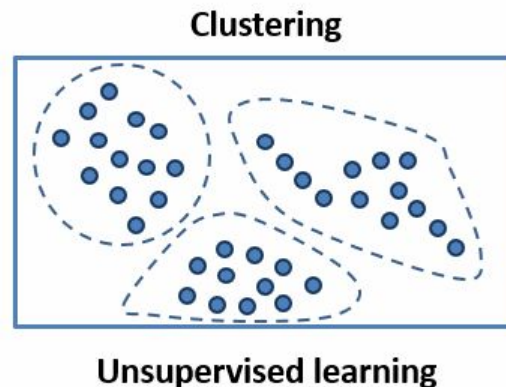
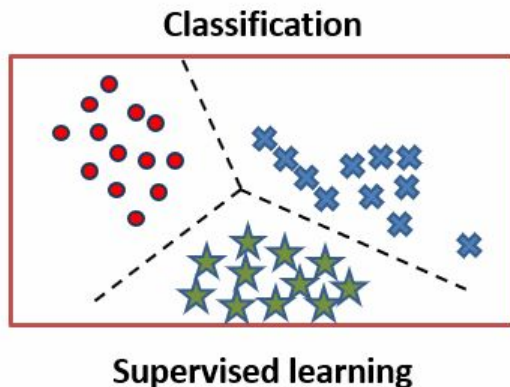
- Clustering is an **unsupervised learning** method that seeks to discover natural patterns in data and dividing them into clusters.
- Smaller clusters are meaningful and useful for understanding complex business problems
- Clustering reduces data dimensions and allows focused analysis of smaller homogenous groups of data



Clustering vs Classification

- The terms “clustering”, “classification” are often used interchangeably.
- **Classification** is grouping observations into some **known categories**. The target category (e.g. fraud) is given upfront.
- **Clustering** is grouping observations into **some unknown categories**. The aim is to find homogenous subsets of observations and group them into clusters.

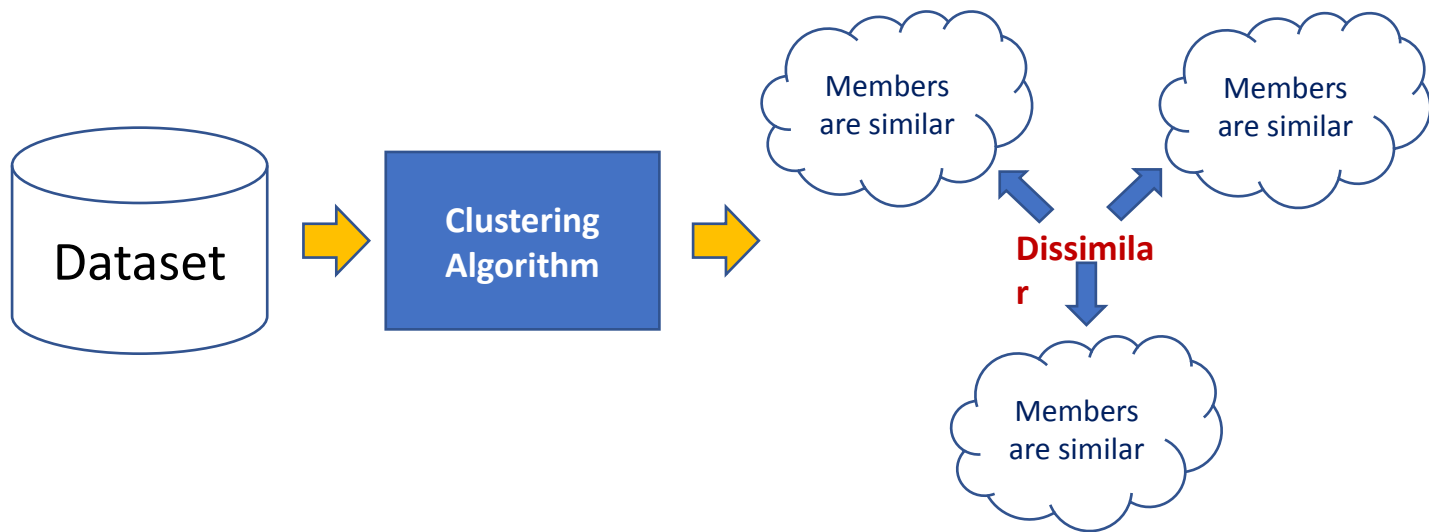
Given some training patterns from each class, the goal is to **construct decision boundaries** within the feature space



Given some data patterns, the goal is to **discover the underlying structure** in the data based on inter-pattern similarities

Overview of the Clustering Technique

- Clustering aims to group data into clusters such that:
 - Observations **within** a cluster are **similar** to each other but...
 - Observations **between** clusters are **dissimilar** to each other



An Example of Clustering

- The World Health Organisation (WHO) would like to identify countries that would benefit from their ***Medical Aid Fund***.
- They want to make that decision based on the **overall country development stage**.
- For example, it would be desirable to have countries grouped into: “Advance”, “Moderate” or “Poor”.
- Countries that are “Poor” will be given more funds (\$) than countries that are “Advance”.



WHO Clustering Objective

WHO would like to find **meaningful groupings** of countries based on the following characteristics:

- *Literacy Rates,*
- *Baby Mortality Rates,*
- *Birth Rates,*
- *Death Rates.*

SN	Country	Literacy	Baby Mort	Birth Rate	Death Rate
1	Argentina	95	25.6	20	9
2	Australia	100	7.3	15	8
3	Bolivia	78	75	34	9
4	Cameroon	54	77	41	12
5	Chile	93	14.6	23	6
6	China	78	52	21	7
7	Costa Rica	93	11	26	4
8	Egypt	48	76.4	29	9
9	Ethiopia	24	110	45	14
10	Greece	93	8.2	10	10
11	Haiti	53	109	40	19
12	India	52	79	29	10
13	Indonesia	77	68	24	9
14	Italy	97	7.6	11	10
15	Kenya	69	74	42	11
16	Kuwait	73	12.5	28	2
17	Mexico	87	35	28	5
18	Nicaragua	57	52.5	35	7
19	Nigeria	51	75	44	12
20	Phillippines	90	51	27	7
21	Somalia	24	126	46	13
22	Thailand	93	37	19	6
23	USA	97	8.1	15	9
24	Vietnam	88	46	27	8
25	Zambia	73	85	46	18

WHO Clustering Objective (cont'd)

A desirable outcome for WHO is the discovery of several clusters of countries such that:

- each cluster has **similar characteristics** but ...
- every cluster is **distinctly different** from each other

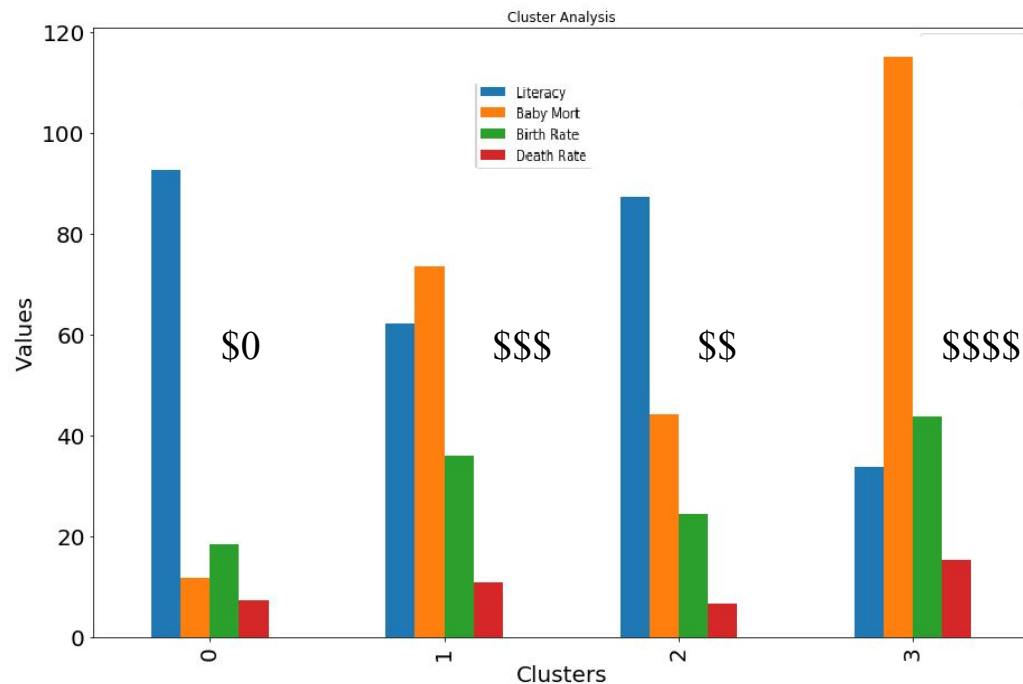
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6		52	21	7	
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8	Egypt	48		9	
9	Ethiopia	24		14	
10	Greece	93		10	
11	Haiti	53		19	
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13	Ind		68	24	9
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15			74	42	11
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17	Mex	87	35	28	5
18	Nicaragua	57	57	7	
19	Nigeria	51		12	
20	Phillippines	90		7	
21	Somalia	24		13	
22	Thailand	93	37	9	6
23	USA	97	8.1	15	9
24	Vietnam	88	46	27	8
25	Zambia	73	85	46	18

Example of Clusters found in the WHO dataset

Results of the Cluster Analysis with 4 clusters

Cluster	0	1	2	3
Literacy	92.625	62.111	87.2	33.667
Baby Mort	11.862	73.544	44.2	115.000
Birth Rate	18.500	36.000	24.4	43.667
Death Rate	7.250	10.778	6.6	15.333
Count	8.000	9.000	5.0	3.000

Country	Nigeria	Vietnam	Haiti
Argentina	Nicaragua	Mexico	Ethiopia
Australia	Kenya	China	Somalia
USA	Indonesia	Phillippines	
Chile	Egypt	Thailand	
Costa Rica	Cameroon		
Kuwait	Bolivia		
Greece	India		
Italy	Zambia		



Did WHO succeed in its business objective?

Clustering in the Business Context

1. Customer Segmentation

- **Purpose:** Group customers with similar behaviors and preferences to enable targeted marketing and improve retention.
- **Real-Life Example: Amazon.com** uses clustering to segment customers based on purchase history, browsing behavior, and demographics.
- **How it works:** Customers who frequently buy tech gadgets are grouped together and shown promotions for new electronics.
- **Impact:** Personalized recommendations and targeted email campaigns increase conversion rates and customer loyalty.

2. Market Research

- **Purpose:** Discover hidden patterns in survey or sales data to understand market segments and design better products.
- **Real-Life Example: Nielsen** applies clustering to consumer survey data to identify emerging lifestyle segments.
- **How it works:** Survey responses are grouped into clusters like “eco-conscious shoppers” or “tech-savvy millennials.”
- **Impact:** Brands use these insights to tailor product features and advertising strategies.

3. Fraud Detection

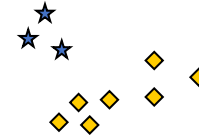
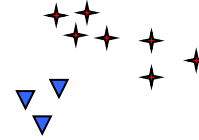
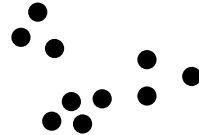
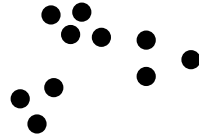
- **Purpose:** Spot unusual behavior that deviates from normal patterns to reduce financial and reputational risk.
- **Real-Life Example:** **PayPal** uses clustering to detect fraudulent transactions.
- **How it works:** Transactions are clustered based on amount, frequency, location, and device. Outliers are flagged for review.
- **Impact:** Early detection of fraud saves millions and protects user trust.

4. Recommendation Systems

- **Purpose:** Group similar users or products to improve suggestions and increase engagement.
- **Real-Life Example:** **Netflix** clusters users based on viewing habits to recommend shows.
- **How it works:** If you watch sci-fi and thrillers, you're grouped with similar users and shown trending content in those genres.
- **Impact:** Personalized recommendations boost watch time and subscriber retention.

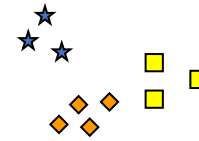
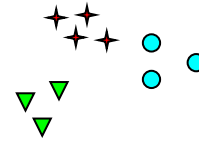
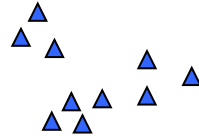
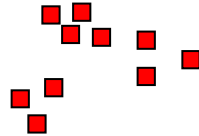
Basic Clustering Concepts: Similarity Measures Cluster Centroids

Different ways of clustering a same set of points



How many
clusters?

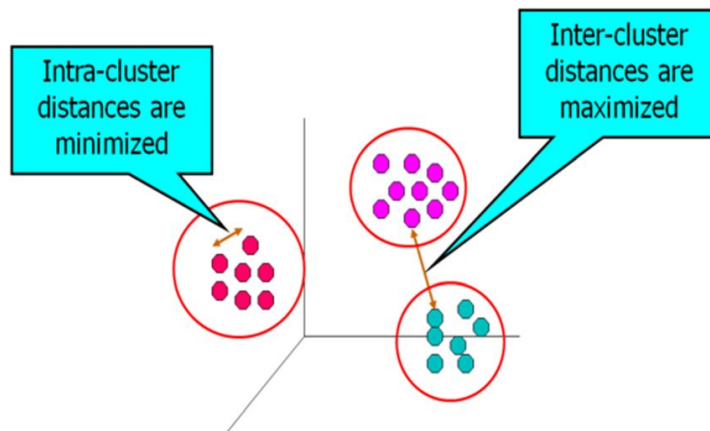
Four Clusters



Two Clusters

Six Clusters

- Clustering algorithms use various **distance/similarity measures** to develop the different clusters.
- A similarity measure is used to decide whether data points are far apart or near to each other.



The goal of clustering is to minimize the Intra-Cluster distance but maximize the Inter-Cluster distance.

Euclidean Distance Measure

- Euclidean Distance is default distance measure for K-means
- Given two data points *i* and *j* (in the p-dimensional space) the Euclidean distance is defined by:

$$d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}$$

Country	Literacy	Baby Mort	Birth Rate	Death Rate
Australia	100	7.3	15	8
Italy	97	7.6	11	10
Somalia	24	126	46	13

$$\sqrt{(100-97)^2 + (7.3-7.6)^2 + (15-11)^2 + (8-10)^2} = 5.3935$$

$$\sqrt{(100-24)^2 + (7.3-126)^2 + (15-46)^2 + (8-13)^2} = 144.4011$$

$$\sqrt{(97-24)^2 + (7.6-126)^2 + (11-46)^2 + (10-13)^2} = 143.4627$$

Australia – Italy: 5.3935

Australia – Somalia:

144.4011

Italy – Somalia:

143.4627

Standardizing your data

Non-Standardized

SN	Age	Income	Dist	
c1	21	5600	1000.61	c1-c2
c2	56	4600	2400.06	c2-c3
c3	39	7000	1400.12	c1-c3

Which pair is closest to each other?

Standardized

SN	Age(S)	Income(S)	Dist	
c1	-1.24	-0.14	2.65	c1-c2
c2	1.21	-1.15	2.71	c2-c3
c3	0.02	1.29	1.90	c1-c3

Which pair is closest to each other?

Ordinal Data (Manhattan Distance)

- Customer Satisfaction: {*Poor*, *Fair*, *Good*, *Excellent*}
- Convert categories to integer indices {1, 2, 3, 4}
- Calculate the absolute difference between indices

Nominal Data (Hamming Distance)

- Job: {Lawyer, Doctor, Teacher}
- Typical approach is:
 - Match = 0 distance
 - Mismatch = 1 distance

SN	Satisfaction	Job
1	Poor	lawyer
2	Excellent	doctor
3	Fair	lawyer

$\text{Dist}(\text{Poor}, \text{Excellent}) = |1-4| = 3$

$\text{Dist}(\text{Poor}, \text{Fair}) = |1-2| = 1$

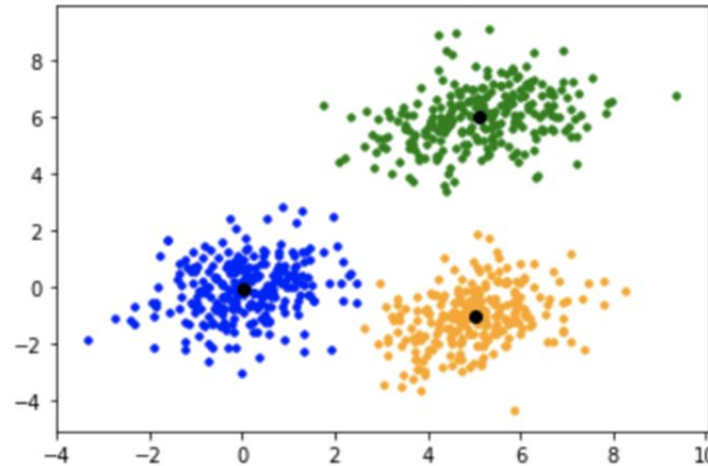
SN	Satisfaction	Job
1	Poor	lawyer
2	Excellent	doctor
3	Fair	lawyer

$\text{Dist}(\text{Lawyer}, \text{doctor}) = 1$

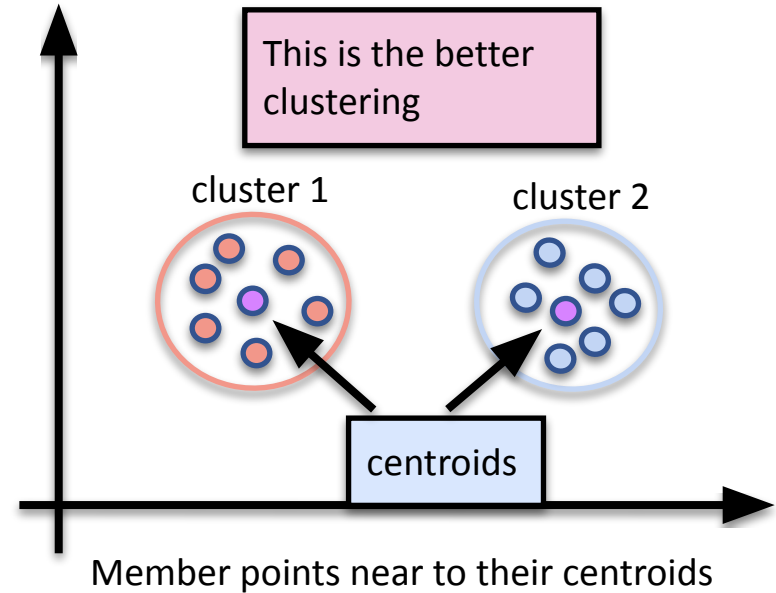
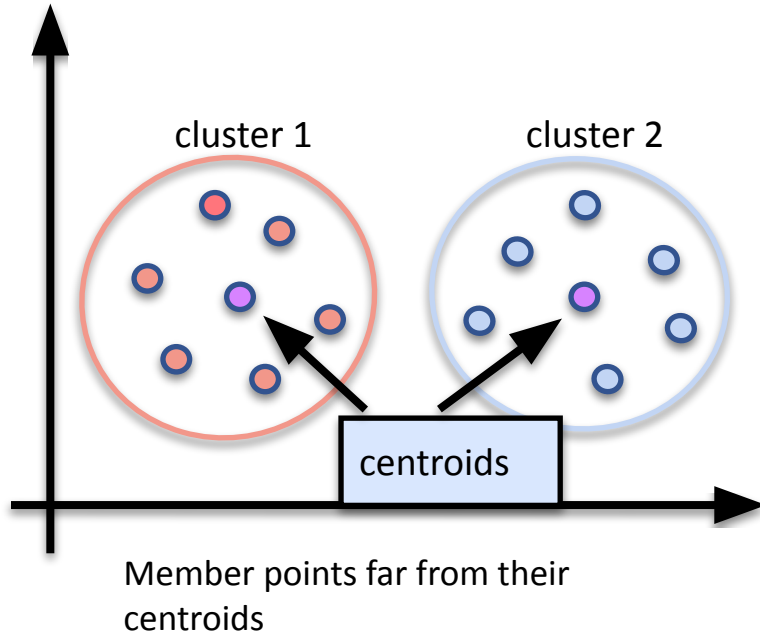
$\text{Dist}(\text{lawyer}, \text{lawyer}) = 0$

Concept of Cluster Centroid

- A **centroid** is the central point of a cluster, typically calculated as the **mean of all data points** within that cluster.
- It serves as a representative location for the cluster and is used to assign new data points based on proximity.
- There are different types of Cluster Centroids depending on the algorithm used: Mean, Medoid, Mode

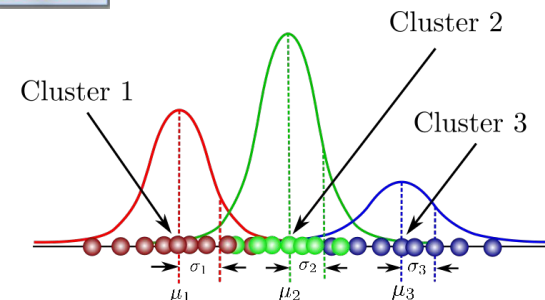
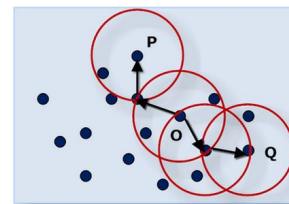
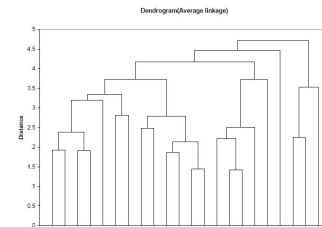
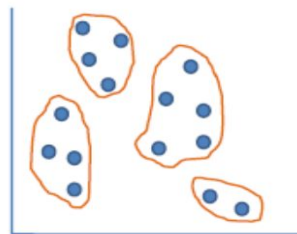


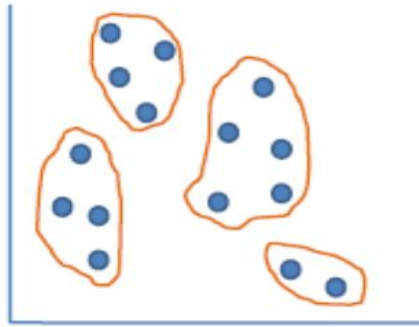
Distance from Centroids



Types of Clustering Approach

- Partition Clustering
 - K-Means Clustering
 - K-modes, k-prototypes
- Hierarchical Clustering
 - Agglomerative
 - Divisive
- Density-based Clustering
 - DBSCAN, HDBSCAN
 - Mean-Shift
- Model-based Clustering (Statistical)
 - Gaussian Mixture Model
 - Expectation-maximization (EM) algorithm





K-Means Clustering

- K-means clustering is a type of top-down divisive clustering
- Requires the user to specify in advance the number of clusters to be formed (k clusters).
- Data observations are then clustered based on its proximity to a nearest cluster centroid.
- **Cluster centroids** are dynamically calculated as cluster members migrate during each iteration of the algorithm.
- It has the advantage of being **more efficient** since it does not generate a complete hierarchy all the way down to individual objects
- k-means is perhaps the **most widely used** clustering algorithm due to its simplicity and efficiency

[https://nlp.stanford.edu/IR-boo
k/](https://nlp.stanford.edu/IR-boo
k/)

K-means clustering algorithm

Specify K number of clusters

Step 1: Arbitrarily designate k data point as **seed points (initial centroids)**

Iterate:

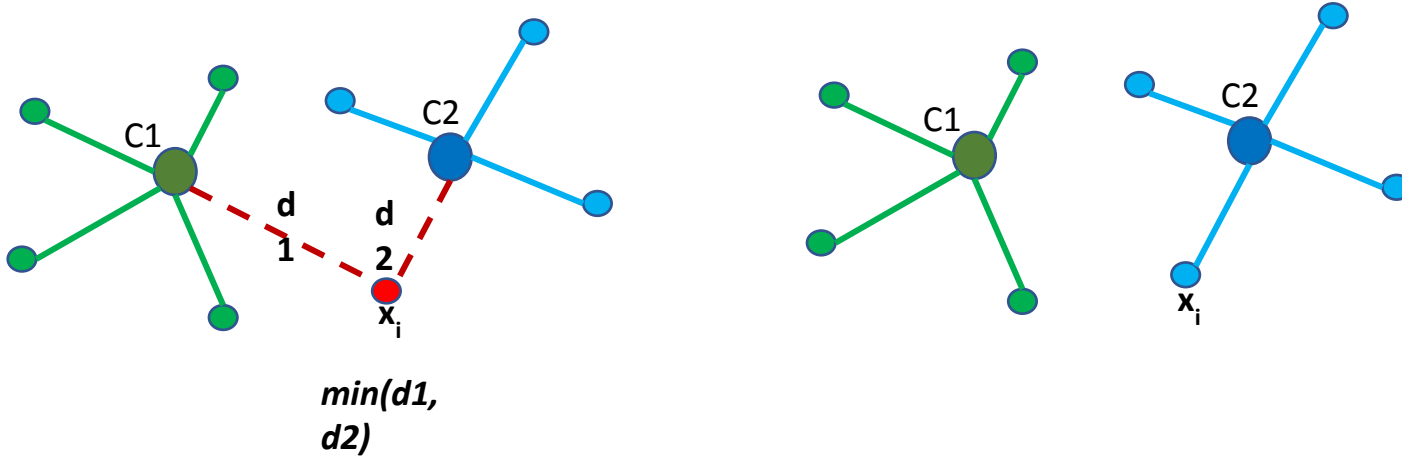
Step 2: **Calculate the distances** between every data point and the centroid.

Step 3: **Re-assign** data points to the **nearest centroid** to form new clusters

Step 4: For each cluster that has lost or gained a data point, **re-compute** the new cluster centroids

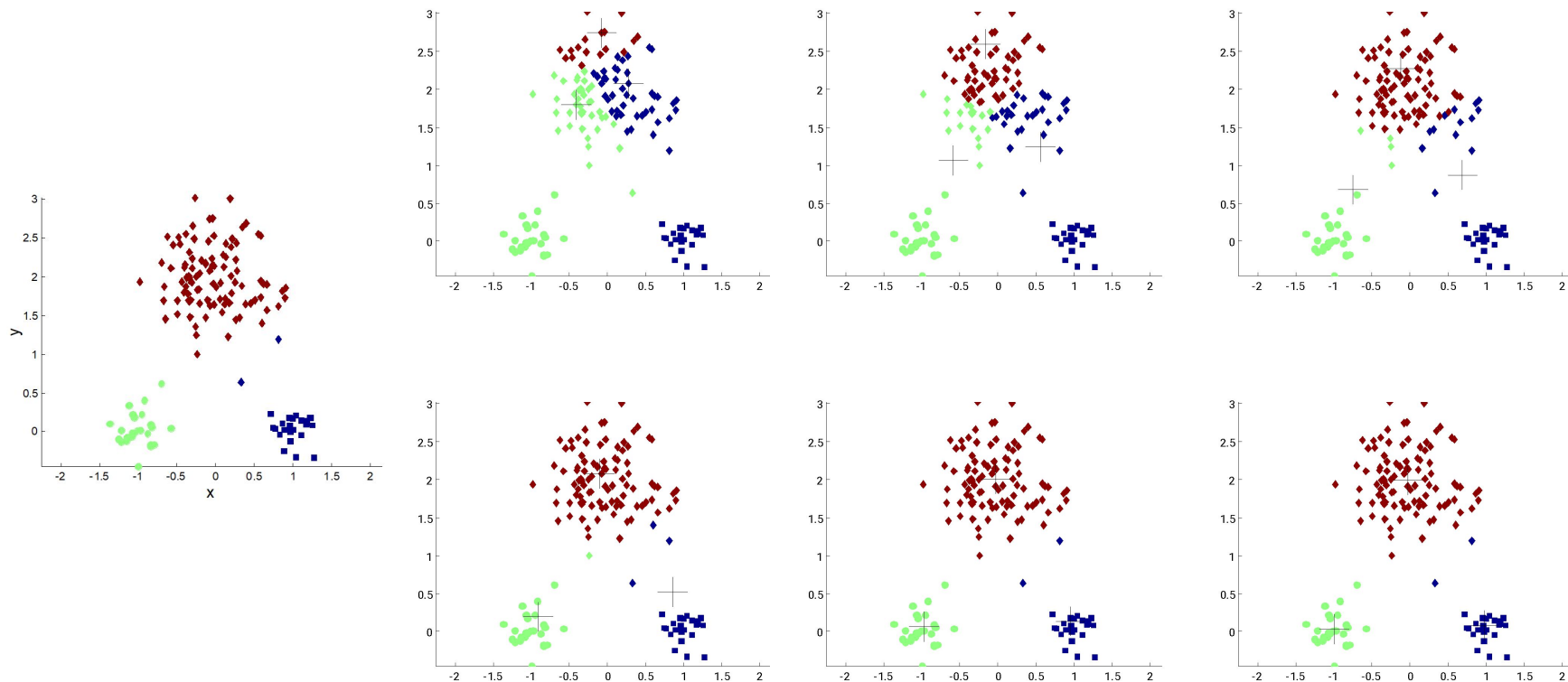
Step 5: Repeat Step 2-4, until (a) no more re-assignment is possible or (b) the newly computed centroids do not deviate significantly from the original or (c) some maximum number of iteration is reached

Assigning data points to nearest centroid



x_i will be assigned to the centroid that is nearer

K-Means clustering method



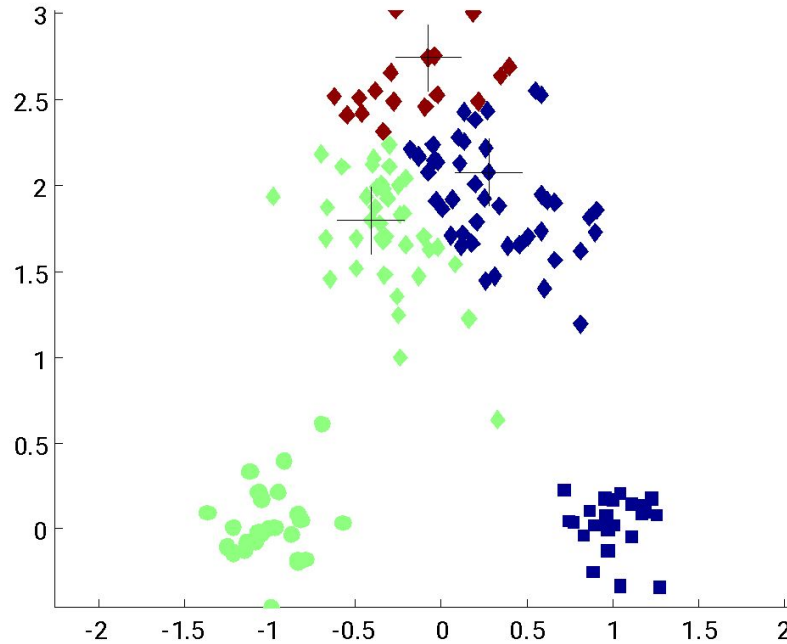
Introduction to Data Mining (Second Edition). Introduction to Data Mining. Pang-Ning Tan, Vipin Kumar

- K-means uses the (Within Cluster) Sum of Squared Error (SSE) as the objective function

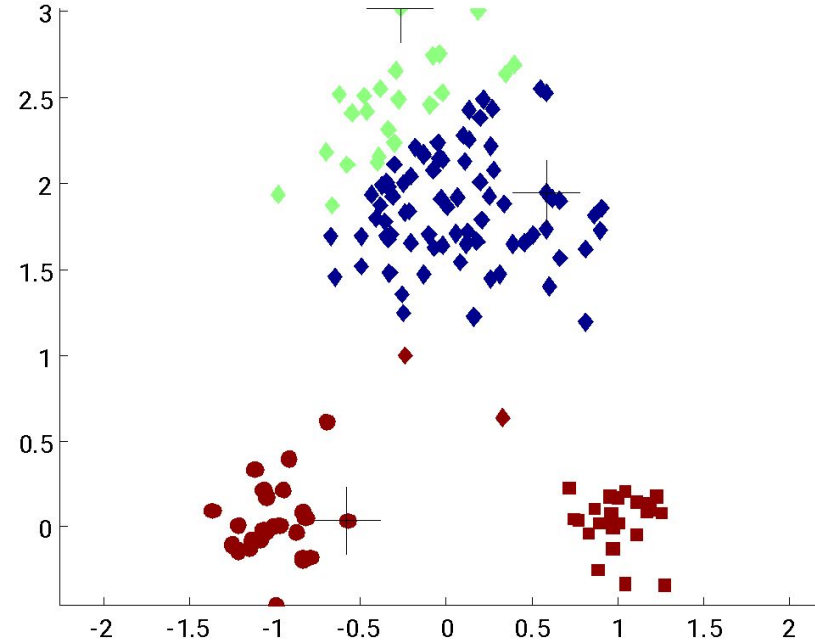
$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- where x is a data point in cluster C_i and m_i is the centroid (mean) for cluster C_i
 - and $dist^2()$ is the Euclidean distance measure
-
- The SSE is expected to improve with each iteration of K-means until it reaches a local or global minima.

Importance of Choosing Initial Centroids ...

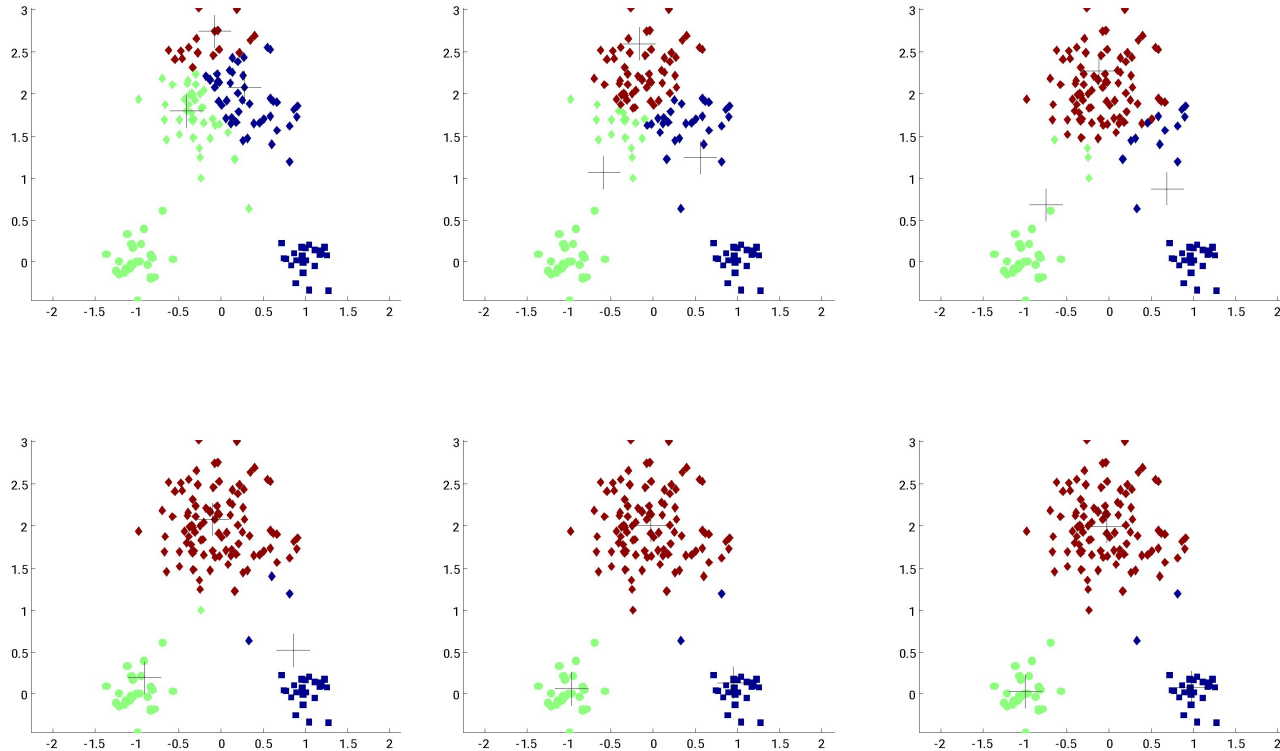


Centroids from the previous example



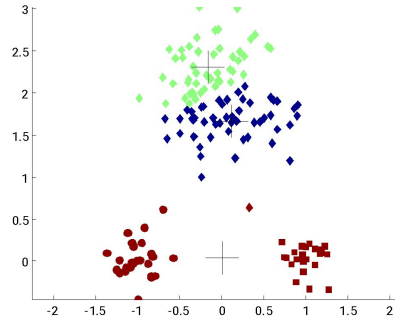
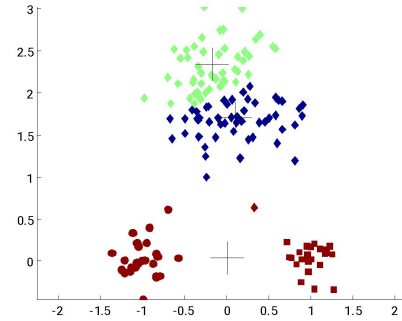
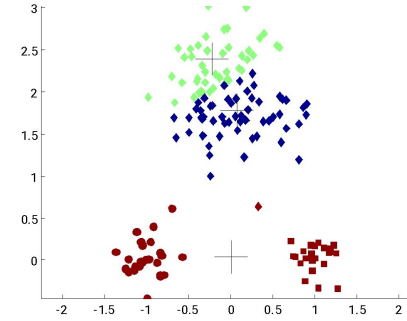
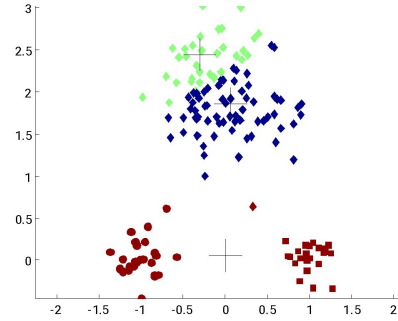
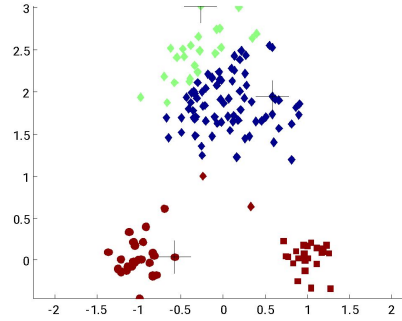
Take a new set of Centroids

Case 1



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Case 2



Introduction to Data Mining (Second Edition). Introduction to Data Mining. Pang-Ning Tan, Vipin Kumar

- Multiple runs (Random Restarts)
 - Run k-means multiple times with different sets of initial random centroids
 - Choose the best result based on the lowest SSE score
 - Increases the likelihood of finding a good solution
 - Tedious and unpredictable
- K-means++ Initialisation
 - Select initial centroids in a way that spreads them out
 - The first centroid is chosen randomly;
 - Subsequent centroids are chosen with probability proportional to their squared distance from the nearest existing centroid
 - Reduces the chances of poor clustering due to bad initial seeds
 - Tendency to include outliers

- A seeding technique by Arthur and Vassilvitskii (2007)
- Idea is to spread the k initial cluster centroids far from each other
- Computationally **more costly** than random initialization but the subsequent k-means often **converges more rapidly**
- Found to improve quality of local optimum and lower runtime

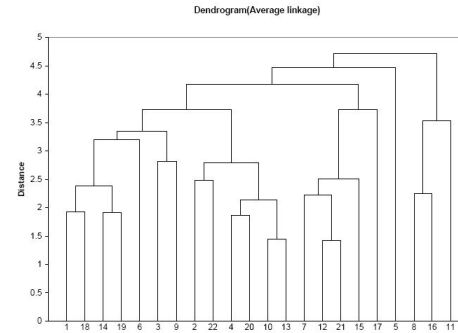
Algorithm 1: k-means++ initialization

```
1:  $C \leftarrow$  sample a point uniformly at random from  $X$ 
2: while  $|C| < k$  do
3:   Sample  $x \in X$  with probability prop. to  $D^2(x)$ 
4:    $C \leftarrow C \cup \{x\}$ 
5: end while
```

Some observations about K-Means

- The K-Means algorithm is guaranteed to converge to a result. However, the result may be a **local optimum** (i.e. not necessarily the best possible).
- Clustering is a **heuristic method** and therefore much experimentation is needed to achieve the desired outcome.
- Pro: Fast, easy, General clustering for all kinds of business problems.
- Con: Sensitive to outlier/noise

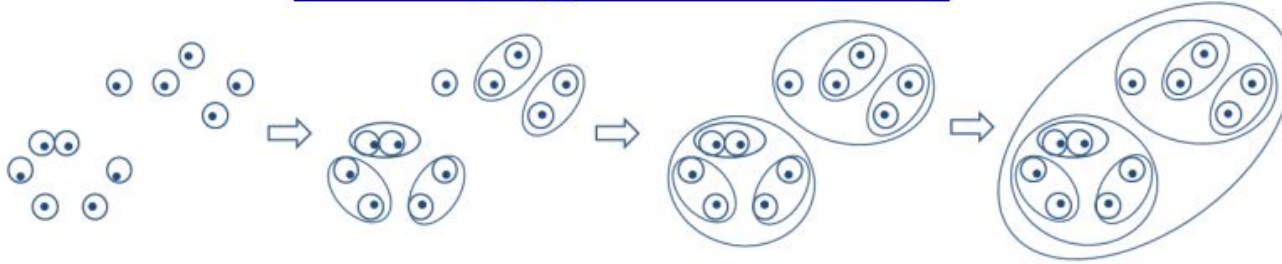
- K-Medoids
 - The medoid of a cluster is defined as the **object** in the cluster whose average dissimilarity to all the objects in the cluster is minimal, that is, it is a most centrally located point in the cluster. (Wikipedia)
- K-Median
 - The median is calculated independently for each feature dimension- so the individual attributes will come from the dataset (or be an average of two values from the dataset). Assign each data point to the cluster whose **median** is closest, using Manhattan distance
- K-modes
 - The K-Modes algorithm is designed specifically for **categorical** data, where K-Means and K-Medians are not suitable due to their reliance on numerical distances. It uses **simple matching** to measure distance and counts how many attributes differ between data points. (<https://pypi.org/project/kmodes/>)
- K-Prototype
 - The k-prototypes algorithm combines k-modes and k-means and is able to cluster **mixed data** (numerical/categorical) types. (<https://pypi.org/project/kmodes/>)



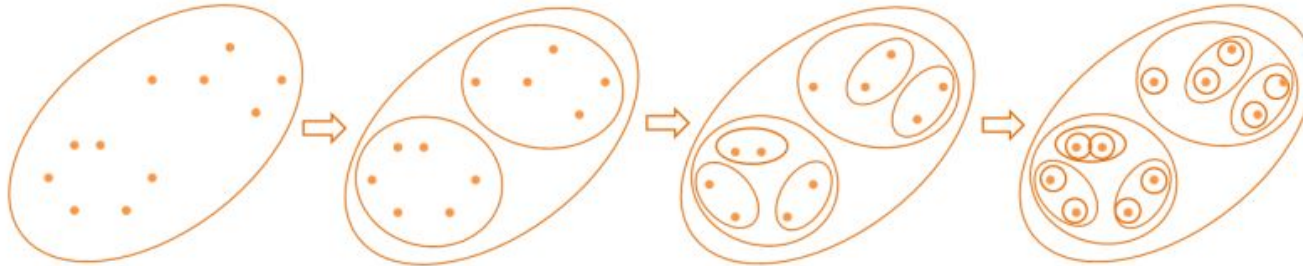
Hierarchical Clustering

Two variants of Hierarchical Clustering

Bottom Up Agglomerative Clustering



Top Down Divisive Clustering



Hierarchical Clustering

Two basic approaches for generating a hierarchical clustering:

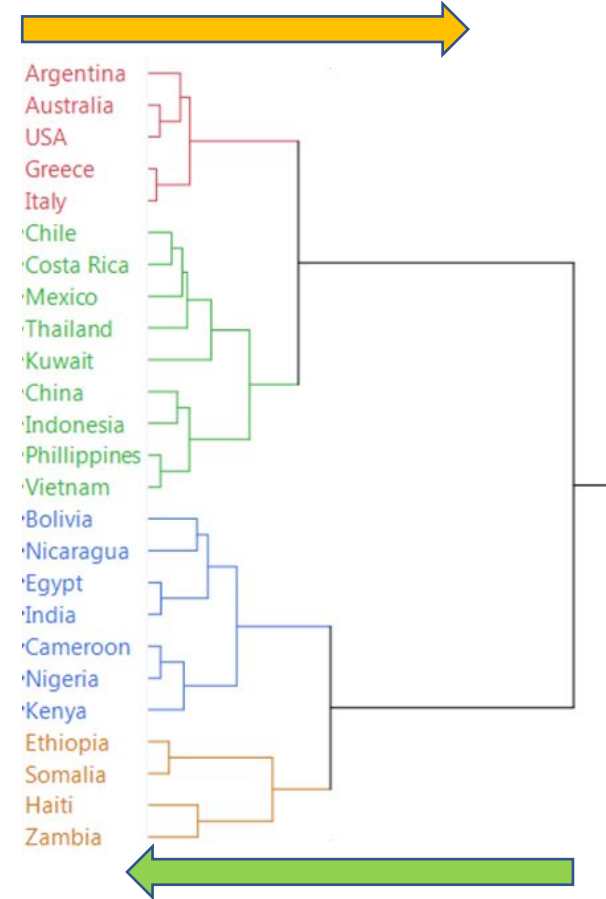
Agglomerative:

- Start with the points as individual clusters and, at each step, merge the closest pair of clusters.
- This requires defining a notion of cluster proximity.

Divisive:

- Start with one, all-inclusive cluster and, at each step, split a cluster until only singleton clusters of individual points remain. In this case, we need to decide which cluster to split at each step and how to do the splitting.

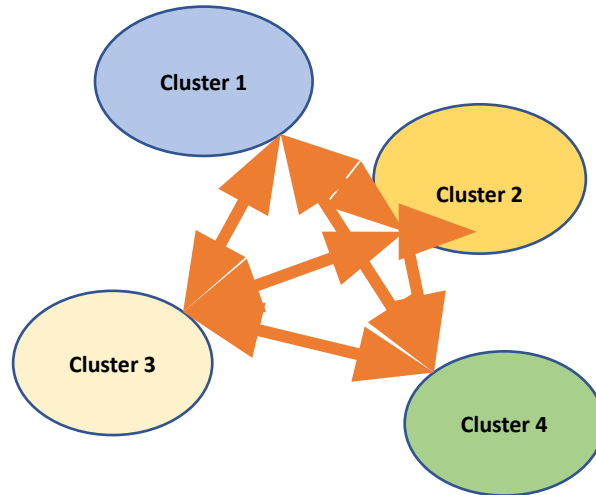
Agglomerative hierarchical clustering is more common than Divisive.



Example

Objective is to find groupings of countries that share similar characteristics in terms of:

- Literacy
- Baby Mortality
- Births
- Deaths



Country	Literacy	Baby Mort	Birth Rate	Death Rate
Argentina	95	25.6	20	9
Australia	100	7.3	15	8
Bolivia	78	75	34	9
Cameroon	54	77	41	12
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Kenya	69	74	42	11
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Mexico	87	35	28	5
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Nigeria	51	75	44	12
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Somalia	24	126	46	13
Thailand	93	37	19	6
USA	97	8.1	15	9
Vietnam	88	46	27	8
Zambia	73	85	46	18

Agglomerative Clustering Algorithm

Start with N data points in the dataset

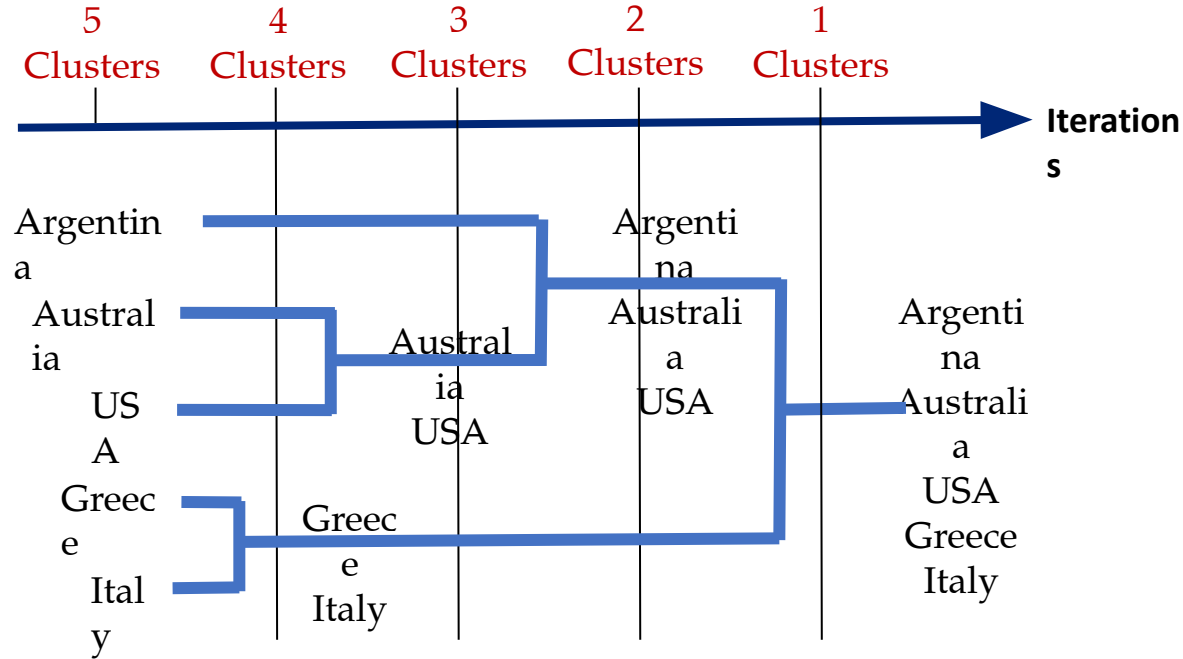
- **Step 1:** Assign each data point as its own cluster

Iterate:

- **Step 2:** Compute the distances between all pairs of clusters
- **Step 3: Merge the two closest clusters** (based on the linkage method).
- **Step 4:** Repeat steps 2-3 until all data points are merged into a single cluster consisting of N members

The results of hierarchical clustering is often displayed graphically using a tree-like diagram called a **dendrogram**, which displays both the cluster-subcluster relationships and the order in which the clusters were merged.

How does Agglomerative Clustering work?



Iteration#1: Greece + Italy
Iteration#2: Australia + USA
Iteration#3: [Australia + USA] + Argentina

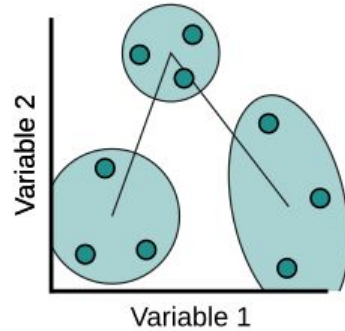
Previously, we learned how to compute distances between objects.
In hierarchical clustering, we need to find **distances between clusters**.

The following are some of these methods:

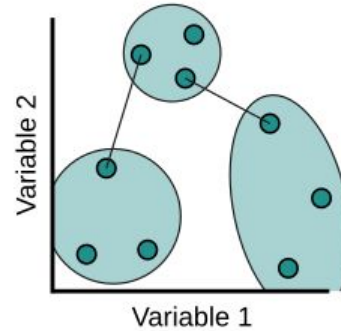
1. Centroid
Distance between the *centroids* of two clusters.
2. Single Linkage
Distance between the *closest points* of two clusters.
3. Complete Linkage
Distance between the *farthest points* of two clusters.
4. Average
Average distance of all points between the two clusters.
5. Wards
Sum of Square error of two clusters if they are merged.

Inter Cluster Distance Measure Methods

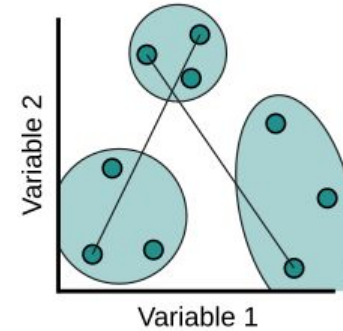
Centroid linkage



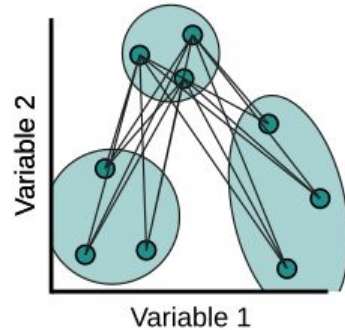
Single linkage



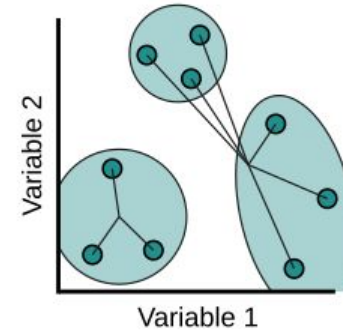
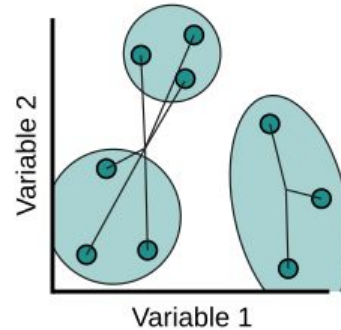
Complete linkage



Average linkage



Ward's method

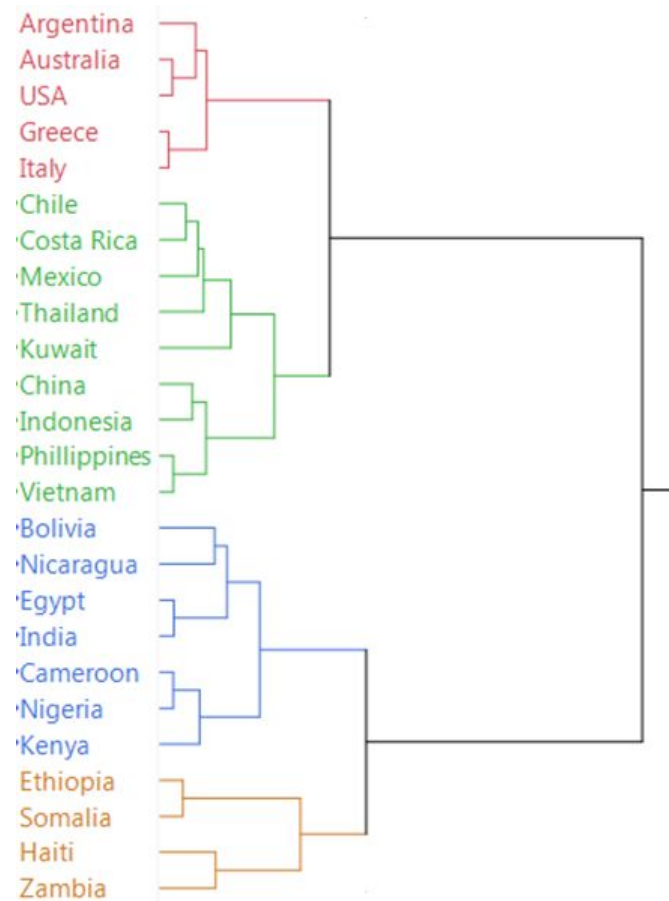


A quick comparison

Linkage Method	Distance Definition	Susceptibility to Outliers	Cluster Shape Tendency
Single	Minimum pairwise distance	High (Chaining effect)	Long, stringy clusters
Complete	Maximum pairwise distance	High (Distorts boundaries)	Tight, compact spherical clusters
Average	Mean of all pairwise distances	Moderate	Balanced, robust clusters
Ward's	Minimizes variance (SSE) increase	Low	Highly uniform, spherical clusters

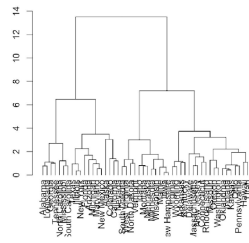
Output of Agglomerative Clustering

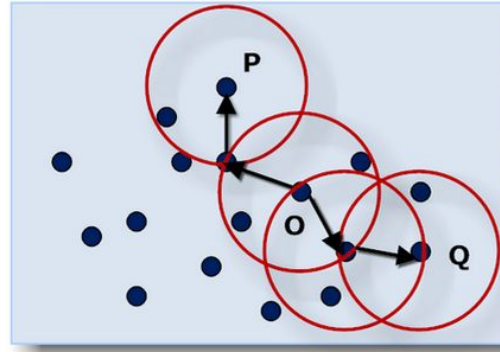
- A **dendrogram** is a tree diagram used to illustrate the arrangement of the clusters
- Read from left to right
 1. Greece + Italy
 2. Australia + USA
 3. Philippines + Vietnam
 4. Cameroon + Nigeria ...
- This color display is for 4 clusters
 - C1: Argentina ... Italy
 - C2: Chile ... Vietnam
 - C3: Bolivia ... Kenya
 - C4: Ethiopia ... Zambia



Limitations of Hierarchical Clustering

- It merges the closest clusters at each step without reconsidering earlier decisions. Once merged, clusters cannot be split again, which can lock in poor choices early on
- Outliers can distort the clustering structure significantly, especially with linkage methods like single linkage.
- It produces a dendrogram, but deciding where to “cut” the tree to form clusters is subjective and can vary depending on the application
- It doesn't scale well with high-dimensional data or large datasets
- Dendrogram can be too large to read for thousands of data observations

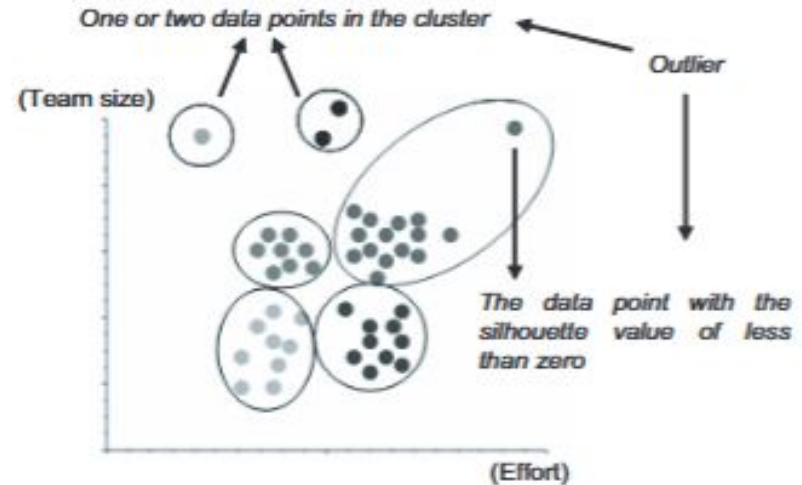




Density-Based Spatial Clustering of Applications with Noise (**DBSCAN**)

Density-Based vs Centroid Based

- K-Means is a **Centroid-based clustering** algorithm because data are assigned to the nearest centroid to form new clusters.
- One of the key strengths of K-Means is that it is easy to understand and efficiently used in practice.
- However, K-Means is highly sensitive to outliers, and can sometimes cause undesirable effects.
- It is possible that outliers are assigned to a nearest cluster where it is not supposed to.

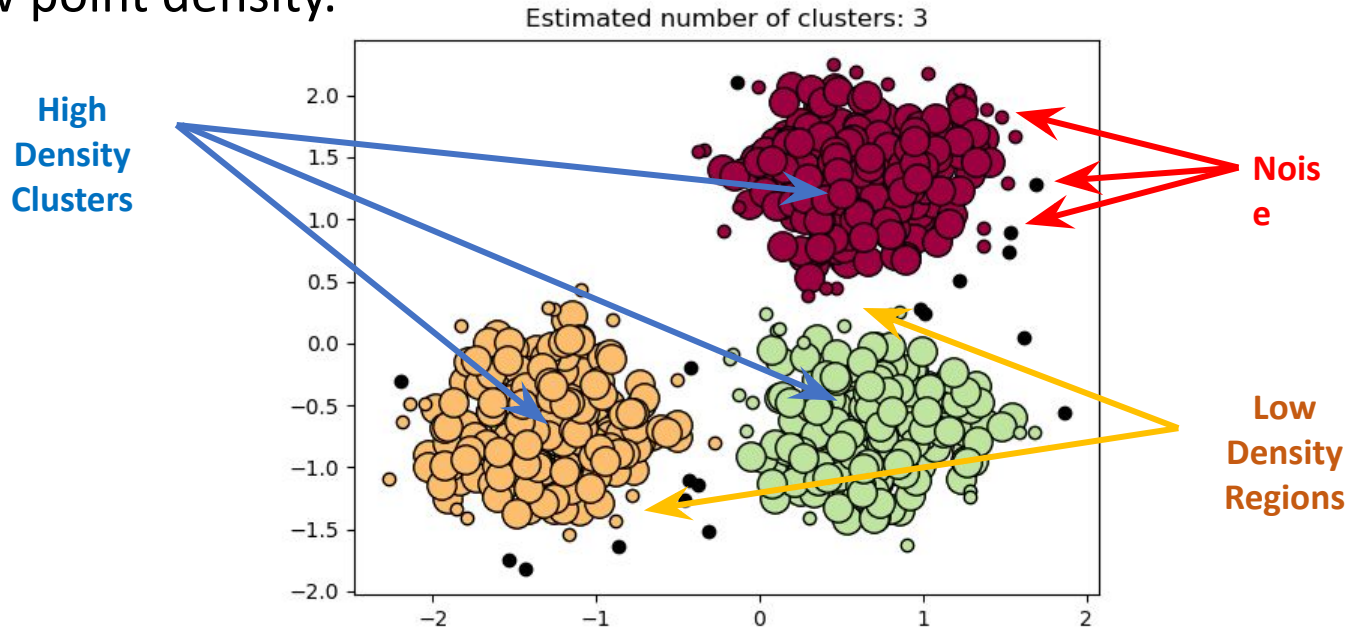


https://www.researchgate.net/figure/An-example-of-outliers-identified-on-Kmeans_fig3_228628775

- In DBSCAN, there are no centroids, and clusters are formed by **linking nearby points** to one another.
- There is also the concept of **noise points** in the DBSCAN algorithm. These are points that do not belong to any clusters. These points are identified by the algorithm.
- There is no need to specify the number of clusters but it does require specifying *two parameters* that influence the decision of whether two nearby points should be linked into the same cluster.
 - A **distance threshold**, Eps (ϵ -epsilon), and
 - A **minimum number of points** (“MinPts”)
- DBSCAN is a **single pass** algorithm. Once a point has been assigned to a particular cluster, it never changes.

How does Density-based clustering work ?

- A clustering method that identifies distinctive clusters in the data, based on the idea that a cluster in data space is a contiguous region of high point density, separated from other such clusters by contiguous regions of low point density.

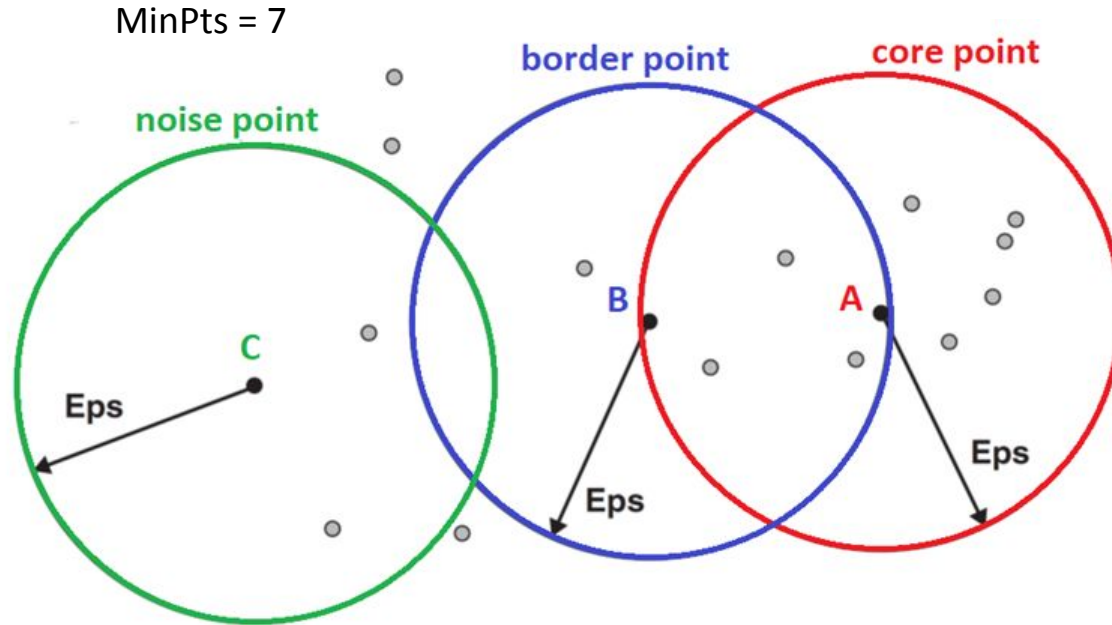


Eps (ϵ -epsilon) : A distance threshold
(The radius around a point to search for neighbors)

MinPts : A minimum number of points required to form a
dense region (Cluster)

- **Density:** at point p = number of points within ϵ
- **Core point:** Has at least MinPts neighbors within ϵ .
 - These are points that are at the interior of a cluster
- **Border point:** Has fewer than **MinPts** neighbours but is within ϵ of a core point
- **Noise point:** is any point that is not a core point or a border point

Core, Border and Noise points



A is core because it has MinPts within its Eps neighborhood

B is a border point because it is within the Eps neighborhood of A but doesn't satisfy MinPts

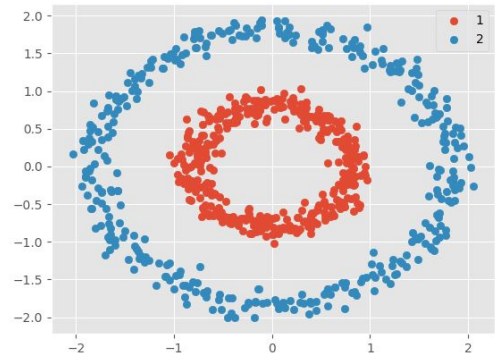
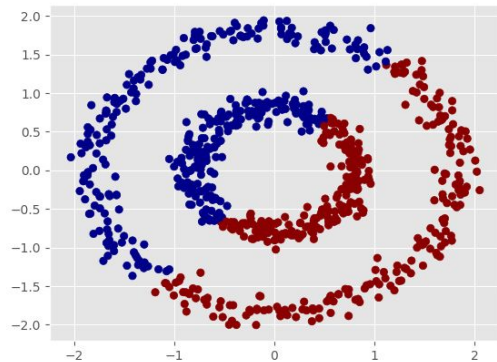
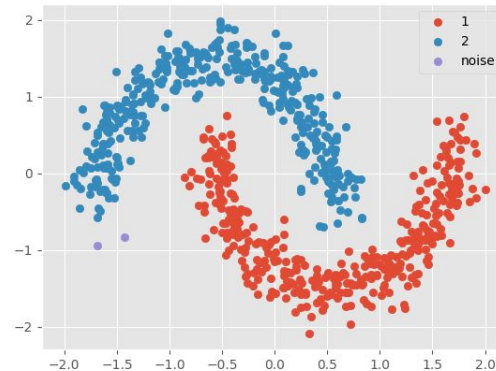
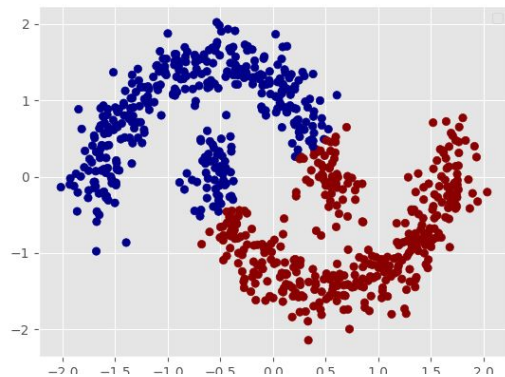
C is a noise point because it is NOT within an Eps neighborhood and have no MinPts

- Pick an unvisited point:
 - If it has $\geq \text{MinPts}$ neighbors within $\epsilon \rightarrow$ it's a core point $\rightarrow$ start a **new cluster**.
 - If not, temporarily mark it as noise (may later be reclassified as border if it falls within ϵ of a core point)
- Expand the cluster:
 - For each **core point**, retrieve all points within its ϵ -neighborhood.
 - Add all **density-reachable** points to the cluster. These include:
 - Other core points (which will recursively expand the cluster)
 - Border Points (which are within ϵ of a core point but are not core points themselves)
 - Continue expanding the cluster until no more density-reachable points can be added
- Repeat:
 - Move to the next unvisited point and repeat the process until all points are either:
 - Assigned to a cluster (as core or border points) or
 - Labeled as noise (not reachable from any core point)
- End: All border points are assigned to a cluster which labeled them as border

Some observations about DBSCAN

- Does not require you to specify the number of clusters
- Can find **arbitrarily shaped cluster** even if the cluster is completely surrounded by a different cluster
- Able to **detect noise** and is robust to outliers
- Pro: Arbitrary shaped data, robust to noise
- Con: require sufficient high-density areas, not popular in business

Kmeans vs DBSCAN



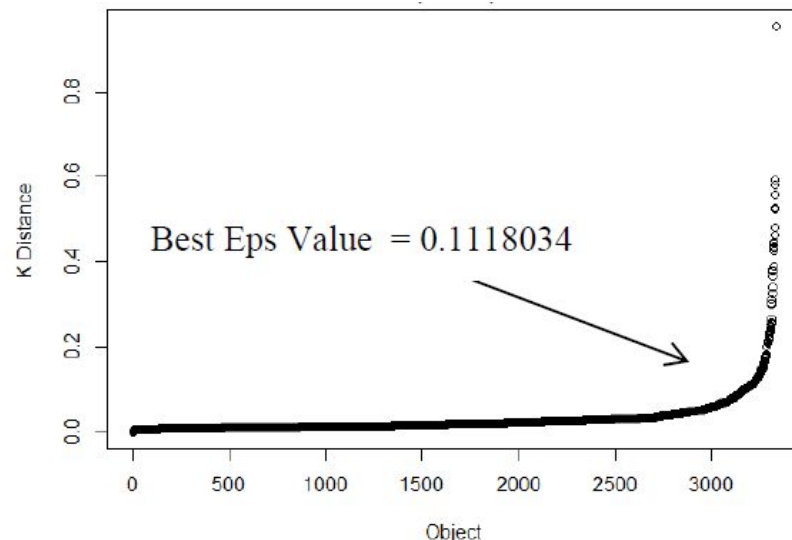
- There is no automatic way to determine the *MinPts* value.
- Ultimately, the *MinPts* value should be set using domain knowledge and familiarity with the data set.
 - The larger the data set, the larger the value of *MinPts* should be
 - If the data set is noisier, choose a larger value of *MinPts*
 - Generally, $MinPts \geq \text{dimensionality of the data set}$
 - For 2-dimensional data, use DBSCAN's default value of $MinPts = 4$ (Ester et al., 1996).
 - If your data has > 2 dimensions, choose $MinPts = 2 * \text{dim}$, where dim= the dimensions of your data set (Sander et al., 1998).

Sander, Jörg; Ester, Martin; Kriegel, Hans-Peter; Xu, Xiaowei (1998). "Density-Based Clustering in Spatial Databases: The Algorithm GDBSCAN and Its Applications". *Data Mining and Knowledge Discovery*. Berlin: Springer-Verlag

DBSCAN Parameter Estimation – Epsilon (ϵ)

The MinPts can be used to estimate the ϵ .

1. Choose a value for **MinPts** (usually 4 or more).
2. For each data point, compute the distance to its **k-th nearest neighbor** (where $k = \text{MinPts}$).
3. Sort these distances in ascending order.
4. Plot the sorted distances.
5. Look for the “**elbow**” point in the graph — this is your estimated ϵ .



Points sorted by distance to the 3rd nearest neighbor

Ref: Nadia Rahmah and Imas Sukaesih Sitanggang 2016 IOP Conf. Ser.: Earth Environ. Sci. 31 012012

Rationale:

- The elbow represents a transition from dense regions (clusters) to sparse regions (noise).
- It helps separate core points from noise.

Evaluating the Validity of Clustering Models

- Clustering algorithms are **guaranteed to converge**, and even when they do, the **quality and stability** of the convergence can vary depending on the algorithm and the data.
- **Cluster Validation** are procedures used to evaluate the quality and reliability of clustering results.
- Clusters are meant to uncover patterns and provide insights for decision making, and therefore clusters must be **valid in the eyes of the stakeholders** – sometimes even when cluster statistics does not provide “strong” support.
- Generally, there are TWO approaches:
 1. **Technical**: Examining and Evaluating Cluster solutions
 2. **Business**: Understanding the Clusters through ProfilingNote these two approaches are usually evaluated and considered in tandem.

Evaluating the Validity of Cluster Models

Technical Approach

1. Cluster Silhouette Coefficient

The Cluster Silhouette coefficient is a combined measure to assess both the intra-cluster-variation (*internal **cohesion***) and the inter-cluster-variation (*external **separation***) of a clustering solution.

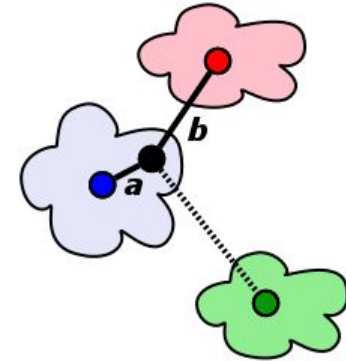
The following statistics is used:-

$$S = \frac{1}{n} \sum_{i=1}^n s(i)$$

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

$a(i)$ = distance between i from its centroid
(*cohesion*)

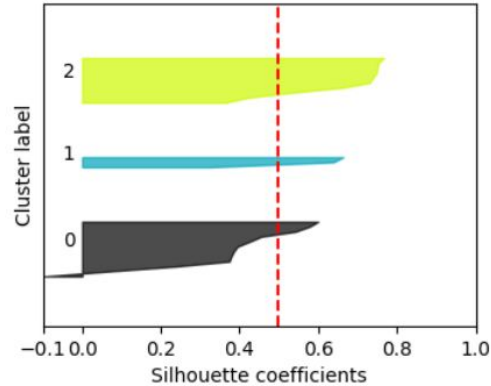
$b(i)$ = distance of i to the nearest cluster
(*separation*)



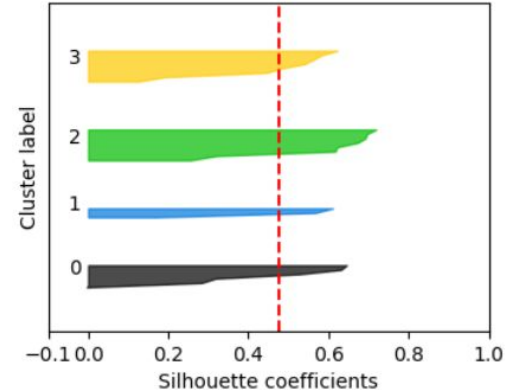
The Silhouette coefficient can vary between -1 (worst) and +1 (best). A score of 0 indicates data lying between clusters. Most tools report that a silhouette score of > 0.3 is acceptable.

Individual Cluster Silhouette Analysis

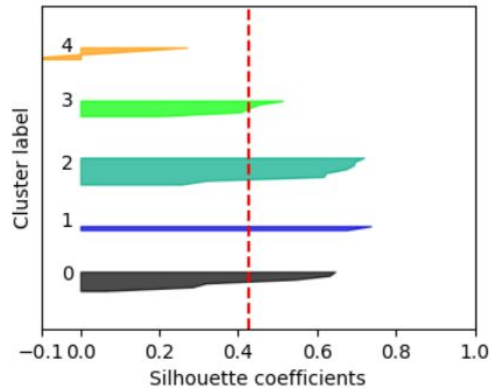
3 Cluster Silhouette



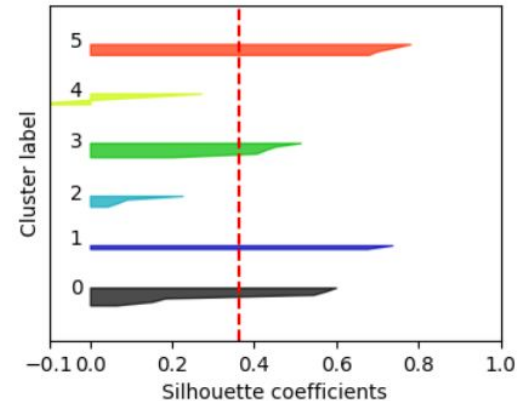
4 Cluster Silhouette



5 Cluster Silhouette



6 Cluster Silhouette



2. Davies–Bouldin Index (BDI)

- Introduced by David L. Davies and Donald W. Bouldin in 1979, the DBI measures how well clusters are separated and how compact they are.
- The DBI is computed as the **average similarity** between each cluster and its most similar one. The similarity is defined as the ratio of:
 - **Intra-cluster distance**: How tightly packed the points are within a cluster.
 - **Inter-cluster distance**: How far apart the cluster centroids are.

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{\Delta_i + \Delta_j}{\delta_{ij}} \right)$$

Lower DBI values indicate better clustering: compact (low intra-cluster distance) and well-separated (high inter-cluster distance).

Higher DBI values suggest overlapping or poorly defined clusters.

- k is the number of clusters.
- Δ_i is the average distance between each point in cluster i and the centroid of cluster i .
- δ_{ij} is the distance between the centroids of clusters i and j
- The maximum is taken over all $j \neq i$, identifying the most similar (least separated) cluster to cluster i

3. Calinski–Harabasz Index (CH)

- The **Calinski–Harabasz Index (CH)**, also known as the **Variance Ratio Criterion (VRC)**. It was introduced in 1974 by Tadeusz Caliński and Jerzy Harabasz
- It measures how well clusters are formed by comparing:
 - **Between-cluster dispersion** (how far apart clusters are)
 - **Within-cluster dispersion** (how compact each cluster is)

$$CH = \frac{BCSS/(k - 1)}{WCSS/(n - k)}$$

- **Higher CHI values** indicate better clustering: Clusters are well-separated (high BCSS) and compact (low WCSS)
- **Lower CHI values** suggest poor clustering structure.

- k = number of clusters
- n = number of data points
- **BCSS** (Between-Cluster Sum of Squares): Measures how distinct the clusters are from each other.
- **WCSS** (Within-Cluster Sum of Squares): Measures how tightly grouped the points are within each cluster

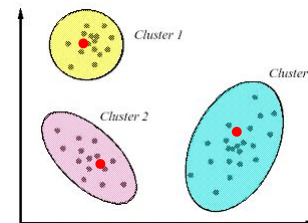
https://en.wikipedia.org/wiki/Calinski%E2%80%93Harabasz_index

4. WCSS Elbow Method

- The basic idea behind k-means clustering consists of defining clusters so that the total intra-cluster (within cluster) variation is minimized.
- Total within-cluster variation is the sum of squared Euclidean distances between data points and the corresponding centroid:

$$WSS = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

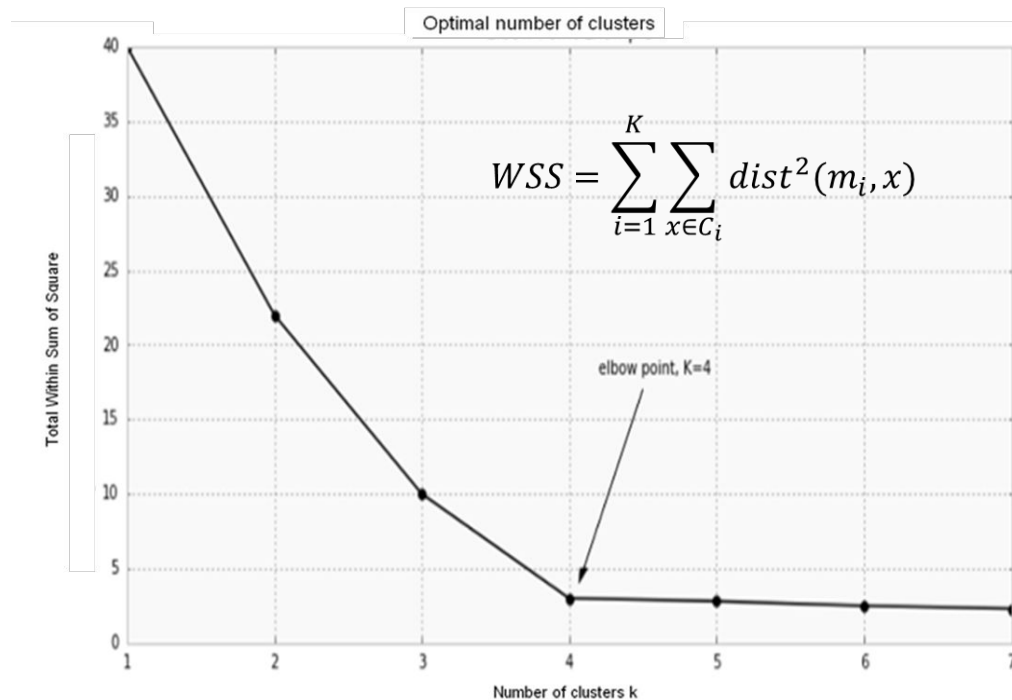
x_i : data point belonging to cluster C_k
 M_k : centroid of cluster C_k



- A good clustering solution is composed of *dense concentration* (compactness) around the centroids. Large dispersion values indicate non-homogenous groupings and suggests further partitioning of the dataset.

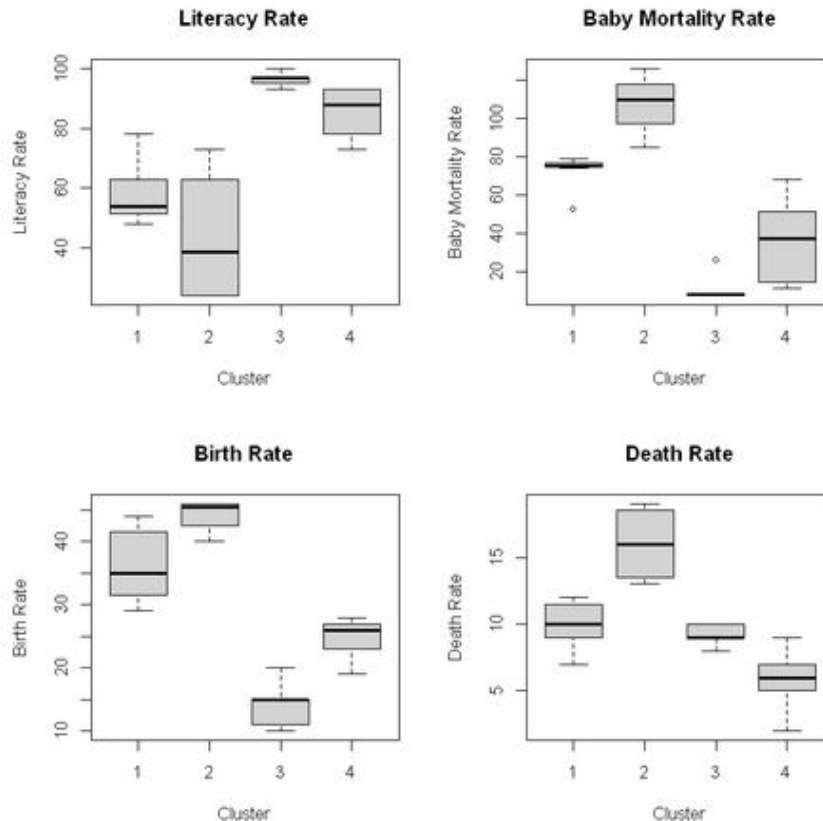
Use WSS to find k

- The Elbow method:
 1. Run k-means for a range of values e.g. $k = 2 \dots 10$
 2. calculate the average WSS for each value of k
 3. Plot a line chart for WSS-vs- K
 4. Look for an elbow - that value of k is the best
- The WSS will get smaller as you increase the number of K clusters.
- Our goal is to choose a small value of k just before (elbow) it starts to show diminishing returns.



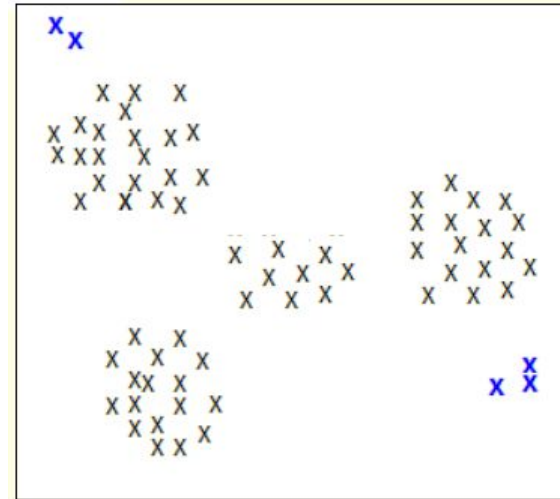
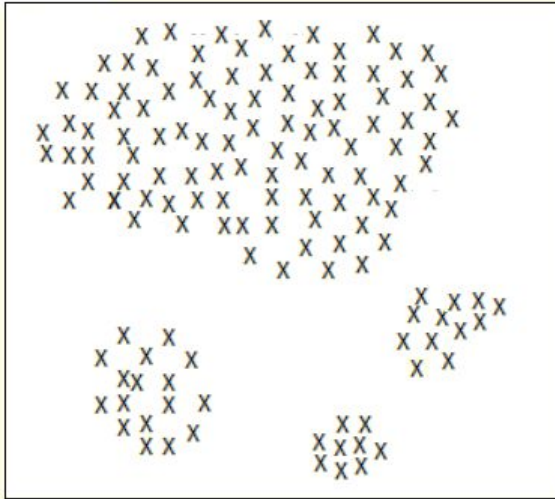
5. Boxplots

- Boxplots offers valuable way to gain insights into the clustering solution and to assess the quality of the clustering results.
- Boxplots are used to show the distribution of each variable for every cluster.
- You will want the data distributions to exhibit significant differences for majority of the variables.
- This plot shows significant differences for Baby Mortality Rates, Birth Rates.



6. Cluster Size

- Examine the **number** of clusters and their relative **sizes**
 - If a single cluster contains majority of the data (i.e. dominates), it could imply the need for further clustering. However, it could also reflect the true profile of your customers
 - If there are very small clusters, it could mean that they represent **outliers**. This should warrant further investigation.

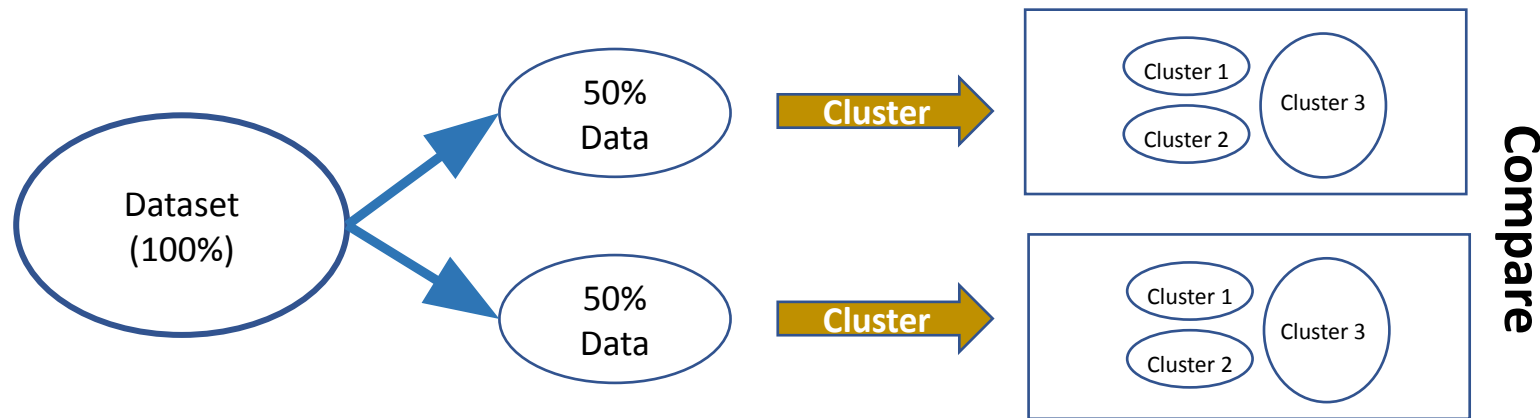


7. Cross Validation method

- Unlike supervised learning, clustering lacks ground truth labels, making validation more complex.
- **Cross-validation for clustering** is a technique used to assess the **stability and reliability** of clustering results by testing how consistent the clusters are across different subsets of the data.
- Cross Validation methods help determine whether the clustering solution is **robust** and **generalizable**.
- Cross-validation for clustering is useful when:
 - You want to ensure your clusters are **not artifacts of random sampling**.
 - You need to **justify clustering decisions** in academic or operational contexts.
 - You're working with **segmentation models** (e.g., student profiles, behavioral clusters) and want to validate their consistency.

7a. Split-half Validation

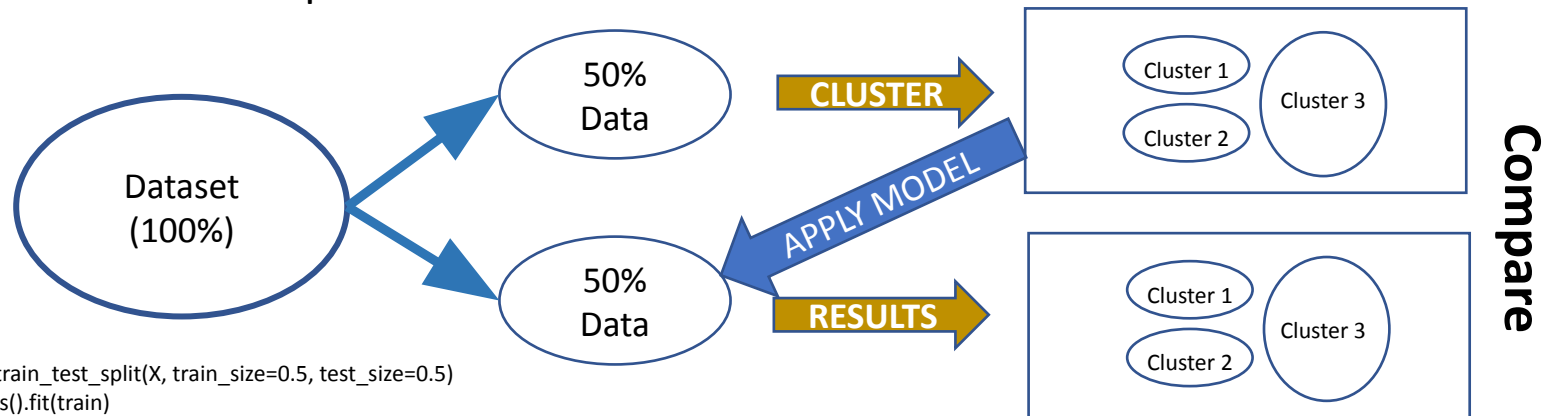
- Randomly split the original data into **two halves**
- Perform clustering separately on each half
- **Compare** the resulting cluster structures
 - Are the number of clusters similar?
 - Do the cluster profiles match?



```
train, test = train_test_split(X, train_size=0.5, test_size=0.5)
km1 = KMeans().fit(train)
km2 = KMeans().fit(test)
```

7b. Development-Validation Split

- Split the dataset into a **development set** and a **validation set** (50:50).
- Perform clustering on the development set.
- Apply the clustering algorithm to the validation set.
- **Compare** the resulting cluster structures
 - Are the number of clusters similar?
 - Do the cluster profiles match?



```
train, test = train_test_split(X, train_size=0.5, test_size=0.5)
km = KMeans().fit(train)
km.predict(test)
```

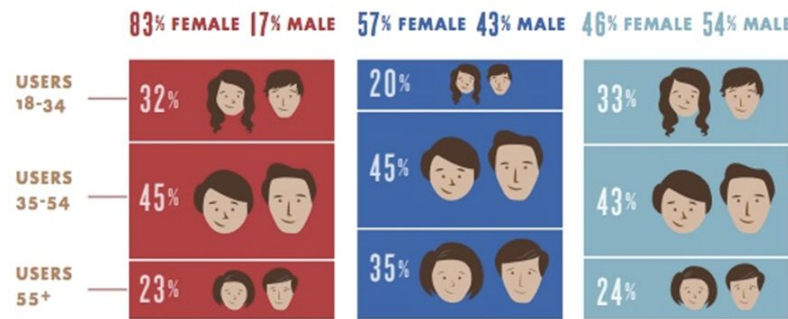
Evaluating the Validity of Cluster Models

Business Approach

Profiling clusters to develop market personas

Understanding cluster solutions through profiling

- Technical measures like the ones discussed earlier are useful but data scientists should also try to understand the **true meaning of each cluster**.
- An understanding of the clusters through profiling is required to take full advantage of them in subsequent marketing activities.
- Profiling requires working closely with the subject matter expert.



Finding the best solution

Interpretability

- Look for *distinguishing characteristics* of each cluster
- Explain each cluster in *practical terms*
- Provide meaningful *labels* for each cluster



Actionability

- Each cluster must be *useful* for marketing, sales or promotion
- Clusters must be able to provide *impetus for action*
- Do you have the *resources* to act?



Potential Business Value

- Clusters must provide *opportunities* to up-sell/cross-sell, or launch new products or services
- *Predict behaviour* of customers and provide better services



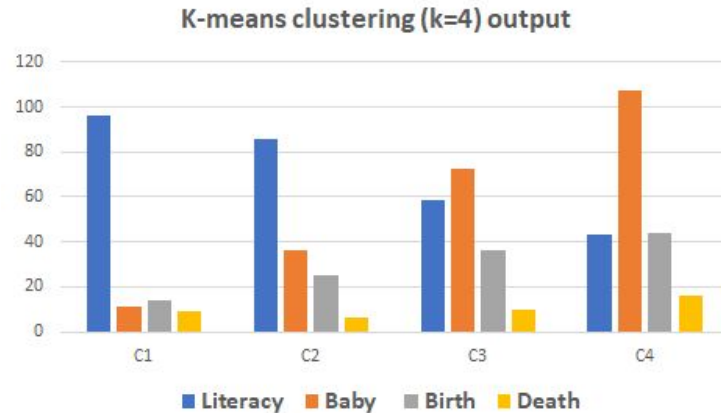
Ease of use

- Clusters must offer *convenience* – eg. it is easier to focus on fewer clusters
- *Align clusters* to your product or service offerings



Develop Profiles from each cluster

- Cluster profiling helps us to gain a *contextual understanding* of the meaning of each cluster.
- Cluster profiling requires the use of *domain knowledge* and *judgement*.
- Criteria to guide cluster profiling:
 1. Look for distinguishing characteristics
 2. Can each cluster be easily explain?
 3. Can labels be assigned intuitively?
 4. Can clusters lead to some action?



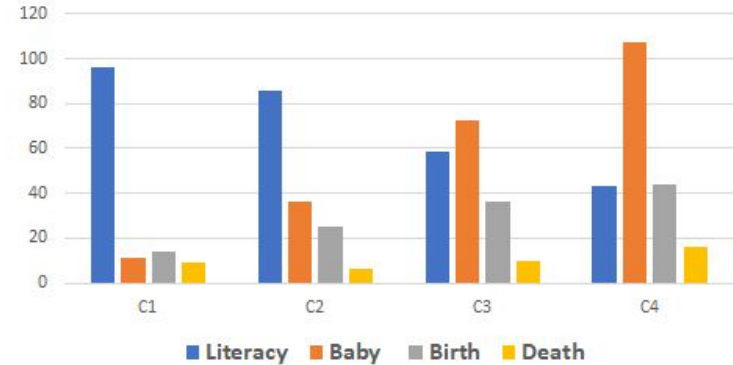
Look for Distinguishing Characteristics

Look for extreme **HIGH** values

Look for extreme **LOW** values

Cluster	Literacy	Baby	Birth	Death	Size
C1	96.40	11.36	14.20	9.2	5.00
C2	85.78	36.34	24.78	6	9.00
C3	58.43	72.70	36.29	10	7.00
C4	43.50	107.50	44.25	16	4.00

K-means clustering (k=4) output



Can each cluster be easily explain?

Can each cluster be explained in the business context?

Cluster	Literacy	Baby	Birth	Death	Size
C1	96.40	11.36	14.20	9.2	5.00
C2	85.78	36.34	24.78	6	9.00
C3	58.43	72.70	36.29	10	7.00
C4	43.50	107.50	44.25	16	4.00

Cluster 1 (High Literacy, Low Baby-Mortality, Birth, Death rates)
appears to be fit the profile of today's advance countries.

Cluster 4 (Low Literacy, High Baby-Mortality, Birth, Death rates)
appears to be fit the profile of today's poorer countries.

Can labels be assigned intuitively?

With the explanation of the clusters, can you assign labels?



Cluster	Literacy	Baby	Birth	Death	Size	Label
C1	96.40	11.36	14.20	9.2	5.00	<i>Advanced</i>
C2	85.78	36.34	24.78	6	9.00	<i>Moderate</i>
C3	58.43	72.70	36.29	10	7.00	<i>Borderline</i>
C4	43.50	107.50	44.25	16	4.00	<i>Poor</i>

Can clusters lead to some action?

With meaningful clusters through labelling, we can also see that actions can be appropriately applied to each cluster

Cluster	Literacy	Baby	Birth	Death	Size	Label
C1	96.40	11.36	14.20	9.2	5.00	<i>Advanced</i>
C2	85.78	36.34	24.78	6	9.00	<i>Moderate</i>
C3	58.43	72.70	36.29	10	7.00	<i>Borderline</i>
C4	43.50	107.50	44.25	16	4.00	<i>Poor</i>

Label	Action
<i>Advanced</i>	No Funding
<i>Moderate</i>	Small Funding (\$)
<i>Borderline</i>	Moderate Funding (\$\$)
<i>Poor</i>	Large Funding (\$\$\$)

1. K-Means Clustering

- **Customer Segmentation in Retail**
- **Use case:** A retail chain segments customers based on purchase frequency and average basket size.
- **Goal:** Identify high-value customers, occasional shoppers, and discount-driven buyers.
- **Outcome:** Tailored marketing campaigns and loyalty programs.

2. Hierarchical Clustering

- **Product Categorization in E-commerce**
- **Use case:** An online marketplace groups products based on descriptions and attributes.
- **Goal:** Automatically organize products into categories and subcategories.
- **Outcome:** Improved search relevance and user navigation.

• 3. DBSCAN

- **Fraud Detection in Financial Transactions**
- **Use case:** A bank uses DBSCAN to detect unusual transaction patterns that deviate from normal customer behavior.
- **Goal:** Flag potentially fraudulent activity without needing labeled data.
- **Outcome:** Reduced false positives and improved fraud investigation efficiency. Automatically detects noise (outliers). No need to predefine the number of clusters.

Demo, Workshop, Exercises

Thank You!